

# Owusu Kantankah

262-302-5015 | [kantankah@wisc.edu](mailto:kantankah@wisc.edu) | [linkedin.com/in/owusukantankah](https://www.linkedin.com/in/owusukantankah) | [github.com/owsu](https://github.com/owsu) | [owusu.info](https://owusu.info)

## EDUCATION

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### University of Wisconsin–Madison

Madison, WI

Bachelor of Science in Computer Science and Data Science

Expected May 2028

- **Coursework:** High Performance Computing (Graduate), Computer Graphics, Algorithms, Databases

## EXPERIENCE

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### LASER Research Program, UW–Madison

Madison, WI

Software Engineer Intern

May 2026 – Aug. 2026

- Built an end-to-end PyTorch/CUDA multi-agent reinforcement-learning system training **100 concurrent agents** across three interchangeable network architectures for controlled architecture experiments.
- Improved training throughput **2.3x** (0.884s → 0.384s per episode) by batching per-group linear layers with `torch.bmm` and reducing GPU launch overhead using `torch.compile` and CUDA graph capture.
- Deployed trained policies to a real-time Three.js client with TensorFlow.js by reimplementing the network forward pass in JavaScript and rendering **100 instanced agents**.
- Received the L&S Summer of Excellence in Research **Rising Researcher Award**; earned a return offer to continue research through the academic year and was selected as a Spring 2027 Computer Graphics TA.

### Space Science and Engineering Center (SSEC)

Madison, WI

Undergraduate Data Engineering Researcher

Sep. 2025 – May 2026

- Built a Python pipeline with NumPy and pandas to process **10GB+** of NetCDF telemetry from JAXA's Akatsuki orbiter, reducing analysis runtime by **35%**.
- Analyzed 1440×720 atmospheric grids to identify a ~195 K polar vortex and 20 K equator-to-pole thermal gradient; presented findings at UW–Madison's Undergraduate Research Symposium.

### StudioNil

Remote

Founder & Software Engineer

Mar. 2023 – Aug. 2025

- Founded and engineered a multiplayer game that reached **10,000 monthly active users in six weeks**.
- Architected client-server infrastructure supporting **100 concurrent players per instance** while maintaining a **0.02% crash rate**.
- Built a custom 3D spatial hitbox system using raycasting, improving collision accuracy by **1.8x** over Roblox's native .Touched API.

## PROJECTS

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### 3D Drone Simulation | *Three.js, GLSL, JavaScript, HTML/CSS* | [GitHub](#)

- Built a 3D drone-swarm simulation implementing Reynolds' boids algorithm across **1,000 agents**, optimized for real-time performance through instanced rendering and spatial hashing.
- Collapsed **1,000 draw calls** into one **InstancedMesh** dispatch, improving frame rate from **80 to 133 FPS** and reducing GPU memory usage to **9 MB**.
- Replaced  $O(n^2)$  pairwise neighbor search with a uniform spatial hash grid using a 27-cell lookup, reducing neighbor queries to near-linear time while keeping the physics loop allocation-free.

### Centralized Morgridge Event Calendar | *Python, Flask, SQLite, AWS* | [GitHub](#)

- Built a RESTful Flask/SQLite API for event retrieval and channel management on a **500-member Discord server**, deployed on AWS EC2 as the data layer between automated scrapers and a React frontend.
- Engineered backend development on a 9-person team, designing the API schema and AWS deployment pipeline for a centralized event calendar serving UW–Madison's **1,200-student** CDIS building.

## TECHNICAL SKILLS

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**Languages:** Python, Java, C++, C, JavaScript, TypeScript, SQL, GLSL, HTML/CSS

**Frameworks/Libraries:** PyTorch, TensorFlow.js, React, Flask, Three.js, NumPy, pandas, Tailwind

**Tools:** CUDA, Docker, AWS (EC2), Linux, CMake, Git, SQLite